STARWEAVER

Hands-On-Learning: Drafting an AI Agent Architecture for Real-Time Threat Prioritization

Instructions

You are a cybersecurity analyst tasked with designing an AI agent to prioritize incoming security alerts in real-time. The agent must collect alerts from a system (e.g., firewall logs), assess their severity using predefined rules and contextual analysis, and prioritize them for human review (e.g., ranking alerts from low to critical). Your goal is to create a high-level architecture diagram that outlines the AI agent's components and their interactions, focusing on how data flows through the system to enable effective decision-making.

Set up your workspace

- Open draw.io (or another free diagramming tool).
- Create a blank diagram.

Review the task context

Imagine you are designing an AI-powered system for a Security Operations Center (SOC) that receives thousands of alerts from IDS/IPS, SIEM, EDR, and other sources.

The goal: prioritize alerts in near-real time, so analysts can focus on the most critical ones.

The AI agent must handle data ingestion, preprocessing, threat scoring, and decision-making.

Draft your architecture

Create a **diagram** that represents the AI agent's high-level architecture.

Include at least the following **core components** (you may add more if needed):

- **Data Ingestion Layer:** Collects alerts from SIEM/IDS/EDR.
- **Preprocessing Module:** Normalization, feature extraction, deduplication.
- **Threat Intelligence Integration:** External feeds, threat scoring context.

AI Agent Core:

- ML Model / LLM Module for classification & prioritization.
- Rule-based Engine for compliance and fallback decisions.

Decision Layer: Produces a risk score / priority level.

Output Interface: Sends prioritized alerts back to SOC dashboards or analysts.

Use arrows to show **data flow** (from raw alerts to final prioritization).

Annotate components with **key technical notes** (e.g., “real-time streaming via Kafka,” “online learning model,” “REST API to SIEM”).

Think about and add some optional components or features to improve the architecture, such as a "Feedback Loop" that adjusts scoring rules based on past human decisions or any other things based on gained knowledge.

Describe key technical considerations

Beneath your diagram (in a text box or as notes), briefly list at least 3 important technical aspects that you might face in case of real deployment scenarios.

Instructions for Documenting and Sharing The Project for Peer Review

Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

Share the Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

Peer Review

Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.